Bivariate Exploratory Data Analysis

Overview

Bivariate Exploratory Data Analysis (EDA) was conducted to investigate the relationships between pairs of variables within the Netflix dataset. Unlike univariate analysis, which focuses on individual variables independently, bivariate analysis examines how two variables interact, enabling the identification of associations, dependencies, and meaningful patterns.

This phase of the analysis addressed two investigation questions:

1. **How do content ratings correlate with specific genres?**
2. **Are certain genres concentrated in specific countries?**

The analysis aimed to understand how Netflix classifies content according to audience maturity across different genres and how genre distribution varies among major content-producing countries. To achieve these objectives, cross-tabulation techniques, percentage distributions, and heatmap visualizations were used to identify significant relationships between the selected variables.

Methodology

The following analytical workflow was adopted for the bivariate analysis:

1. Loaded the cleaned Netflix dataset.
2. Selected the variables required for each investigation question.
3. Removed records containing missing values.
4. Processed multi-valued categorical variables (Type and Country) by splitting comma-separated entries into individual values.
5. Expanded the extracted values using the explode() function to facilitate pairwise analysis.
6. Constructed contingency tables using pd.crosstab() to summarize relationships between variables.
7. Calculated row-wise percentage distributions to enable meaningful comparisons.
8. Visualized relationships using:
    - Frequency Heatmaps
    - Percentage Heatmaps
    - Focused Heatmaps for Top Genres and Top Countries
9. Interpreted observed patterns and derived business insights based on the visualizations.

Investigation Question 5

How do content ratings correlate with specific genres?

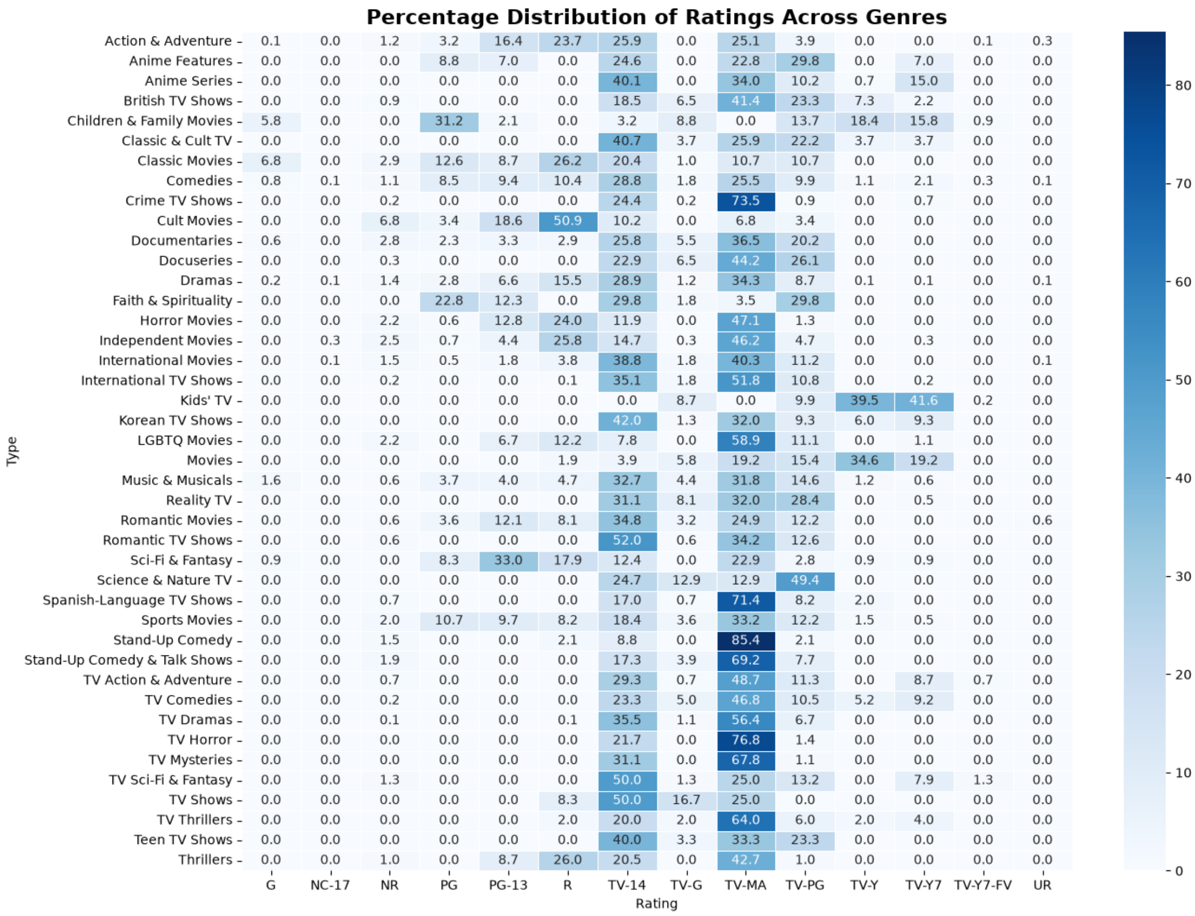
Objective

The objective of this analysis was to investigate how Netflix's content ratings vary across different genres. By examining the distribution of maturity ratings within each genre, the analysis provides insights into the intended audience for different categories of content and Netflix's overall content classification strategy.

Analysis Performed

The following analytical steps were carried out:

- Extracted the **Rating** and **Type** columns.
- Split the multi-valued genre field into individual genres.
- Expanded the dataset using the explode() function.
- Created a cross-tabulation between genres and maturity ratings.
- Calculated percentage distributions within each genre.
- Visualized the relationships using frequency and percentage heatmaps.
- Performed a focused analysis on the top ten most prevalent genres.



Major Findings

- **TV-MA** and **TV-14** are the dominant maturity ratings across most Netflix genres.
- **International Movies**, **Dramas**, and **International TV Shows** are primarily intended for mature audiences.
- **TV Dramas** exhibit the highest concentration of **TV-MA** ratings, indicating strong emphasis on adult-oriented storytelling.
- **Children & Family Movies** are overwhelmingly associated with **G**, **PG**, **TV-Y**, and **TV-Y7** ratings, confirming clear audience segmentation.
- Genres such as **Action & Adventure**, **Documentaries**, and **Comedies** serve multiple audience groups by spanning several maturity classifications.

Interpretation

The findings demonstrate that Netflix aligns content ratings closely with genre characteristics. Mature genres consistently receive higher maturity classifications, while children's content remains appropriately categorized for younger audiences. This structured rating strategy supports effective content discovery, parental guidance, and personalized recommendation systems.

Investigation Question 6

Are certain genres concentrated in specific countries?

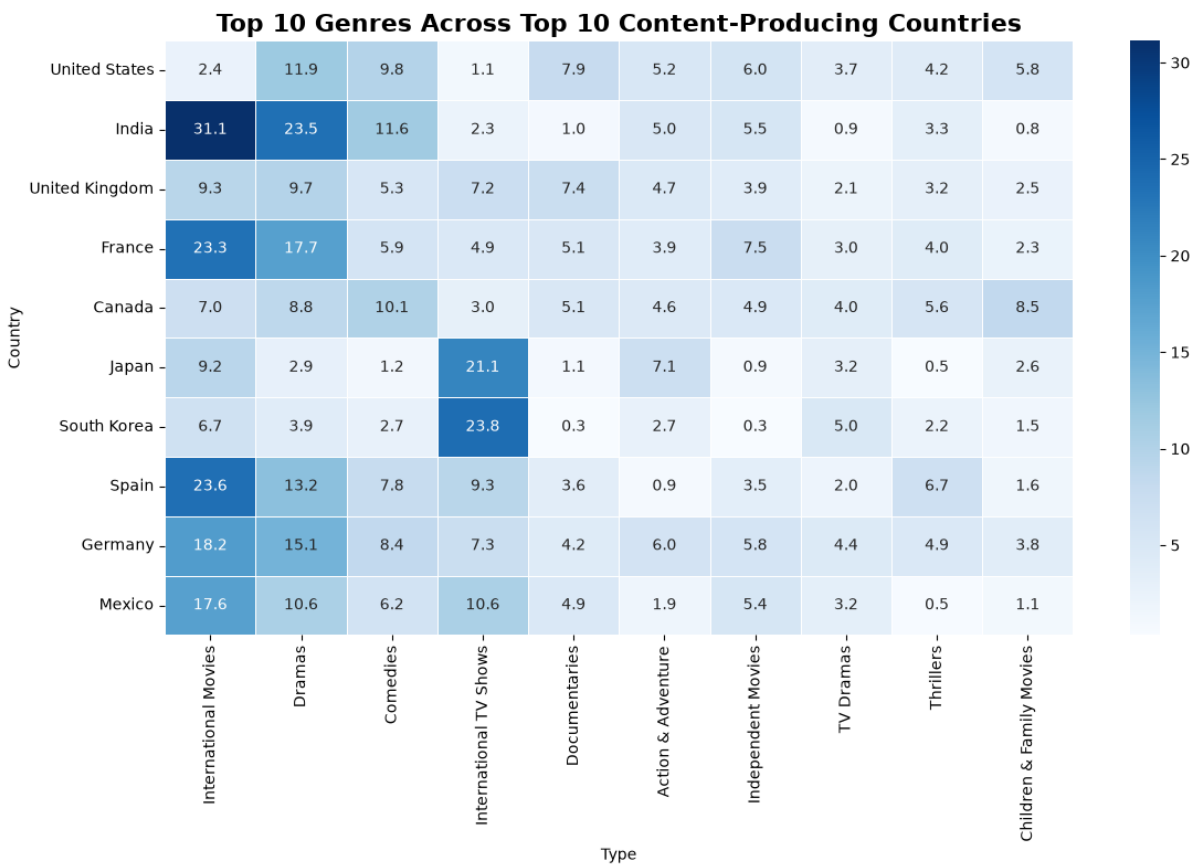
Objective

The objective of this analysis was to investigate whether specific genres are concentrated within particular content-producing countries and to identify regional production specializations within Netflix's global catalogue.

Analysis Performed

The following steps were performed:

- Extracted the **Country** and **Type** columns.
- Split multiple countries and genres into individual observations.
- Expanded the dataset using the explode() function.
- Created cross-tabulations between countries and genres.
- Calculated percentage distributions by country.
- Generated an overall country–genre heatmap.
- Produced a focused heatmap showing the relationship between the **Top 10 Content-Producing Countries** and the **Top 10 Genres**.



Major Findings

- Genre distribution varies considerably across countries, indicating strong regional specialization.
- **India** exhibits the highest concentration of **International Movies (31.1%)**, demonstrating the country's strong focus on film production.
- **Japan (21.1%)** and **South Korea (23.8%)** show the greatest concentration of **International TV Shows**, reflecting their specialization in serialized television content.
- The **United States** maintains the most balanced genre distribution among all major content-producing countries, contributing consistently across dramas, comedies, documentaries, children's content, and independent films.
- **France**, **Spain**, **Germany**, and **Mexico** also demonstrate strong emphasis on **International Movies**, while **Canada** displays one of the most diversified genre portfolios.

Interpretation

The analysis confirms that Netflix leverages the strengths of different regional entertainment industries to build a globally diverse content catalog. Rather than relying on uniform content acquisition, the platform combines regional specialization with globally popular genres, allowing it to appeal to audiences across different cultures and geographical markets.

Overall Findings

The bivariate analysis provided valuable insights into the relationships between Netflix's content ratings, genres, and countries of production.

The key findings are summarized below:

- Content ratings are strongly associated with genre characteristics, with **TV-MA** and **TV-14** dominating the majority of Netflix's catalog.
- Family-oriented genres consistently receive child-appropriate ratings, demonstrating effective audience segmentation.
- Mature genres such as dramas, crime, international productions, and independent films are primarily targeted toward adult viewers.
- Genre distribution differs significantly across countries, indicating that regional production industries specialize in different types of content.
- India primarily contributes internationally distributed films, whereas Japan and South Korea specialize in internationally recognized television productions.
- The United States maintains the most balanced and diversified content portfolio among all major production countries.
- Netflix's global catalog reflects a combination of internationally popular genres and regionally specialized productions, supporting worldwide audience engagement.

Overall, the relationships identified during the bivariate analysis demonstrate that Netflix's content acquisition strategy is both audience-driven and geographically diversified.

Business Insights

The bivariate analysis highlights several strategic characteristics of Netflix's content ecosystem:

- Netflix successfully aligns maturity ratings with genre characteristics, enabling viewers to easily identify age-appropriate content.
- The dominance of mature ratings across major genres reflects the platform's emphasis on serving adult and young adult audiences.
- Regional specialization enables Netflix to leverage the comparative advantages of different entertainment industries while maintaining a globally diverse content catalog.
- The balanced genre portfolio of the United States complements the specialized contributions of countries such as India, Japan, and South Korea, resulting in a richer and more diverse streaming library.
- Strong geographical diversification reduces dependence on a single production market and supports localized content recommendations for international audiences.

Business Recommendations

Based on the findings obtained from the bivariate analysis, the following recommendations are proposed:

1. Continue Investing in Mature Content

The dominance of **TV-MA** and **TV-14** across major genres indicates sustained audience demand for mature storytelling. Continued investment in these categories is likely to strengthen viewer engagement.

2. Expand Family-Oriented Productions

Although children's content is well represented, increasing the availability of family-friendly titles across additional genres could broaden Netflix's appeal to younger audiences and households.

3. Strengthen Regional Content Partnerships

Countries such as India, Japan, South Korea, France, and Spain demonstrate clear production strengths. Expanding partnerships within these markets can further enhance Netflix's international content portfolio.

4. Improve Recommendation Systems

The observed relationships between genres, maturity ratings, and countries can be incorporated into recommendation algorithms to deliver more personalized viewing experiences.

5. Preserve Geographical Diversity

Maintaining a balanced combination of globally popular genres and region-specific productions will continue to strengthen Netflix's competitive advantage in international streaming markets.

Conclusion

The Bivariate Exploratory Data Analysis successfully examined the relationships between content ratings, genres, and countries within the Netflix dataset. The analysis demonstrated that Netflix's maturity ratings closely align with genre characteristics and that genre distributions vary significantly across different content-producing countries.

The findings confirm that Netflix follows a structured content classification system while simultaneously adopting a geographically diversified acquisition strategy that leverages regional production strengths. These relationships illustrate how Netflix balances globally popular entertainment with culturally distinctive content to serve audiences across diverse markets.

Overall, the bivariate analysis provides a deeper understanding of Netflix's content organization, audience targeting strategy, and global content sourcing practices. These insights establish a strong analytical foundation for future predictive modeling, recommendation system development, and strategic business decision-making.